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EE1060: Differential Equations and Transform Techniques

Lecture 24: Fourier Transform

Topics :

1. Properties of Fourier Transform
 2. Convergence of Fourier Transform
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Simple Examples of Fourier Transform

Fourier Transform of $x(t)$ $X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$ (Analysis)

Inverse Fourier Transform $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$ (Synthesis)

FT Pair $x(t) \Leftrightarrow X(\omega)$
 $\xrightarrow{\quad}$ can be considered as a signal function

from Lec 23:- $x(t) = 1 \Leftrightarrow ?$ (Not defined)
 $X(\omega)$ (intuition)

Example:- $x(t) = \delta(t)$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt$$

$$= e^{-j\omega(0)} = 1.$$

$$\int_{-\infty}^{\infty} \delta(t-\tau) f(t) dt = f(\tau)$$

$x(t) = u(t)$

$$X(\omega) = \int_{-\infty}^{\infty} u(t) e^{-j\omega t} dt = \int_0^{\infty} e^{-j\omega t} dt = \frac{1}{-j\omega} e^{-j\omega t} \Big|_0^{\infty}$$

Not defined?

Time shift and Time scaling

$x(t) \Leftrightarrow X(\omega)$. Compute $Y(\omega)$ for $y(t) = x(t-\tau)$.
time shift

$$Y(\omega) = \int_{-\infty}^{\infty} x(t-\tau) e^{-j\omega t} dt$$

$$\text{let } t-\tau = t_1 \Rightarrow Y(\omega) = \int_{-\infty}^{\infty} x(t_1) e^{-j\omega t_1} e^{-j\omega \tau} dt$$

$$= e^{-j\omega \tau} \underline{X(\omega)} \quad \begin{array}{l} \text{mag} = \text{unchanged} \\ \text{Phase} \propto f(\omega) \end{array}$$

$x(t) \Leftrightarrow X(\omega)$. Compute $Y(\omega)$ for $y(t) = x(at)$?

$$Y(\omega) = \int_{-\infty}^{\infty} x(at) e^{-j\omega t} dt$$

$$\text{let } at = t_1$$

$$dt = \frac{1}{|a|} dt_1$$

$$= \frac{1}{|a|} \int_{-\infty}^{\infty} x(t_1) e^{-j\omega t_1/a} dt_1$$

$$= \frac{1}{|a|} \int_{-\infty}^{\infty} x(t_1) \underbrace{e^{-j(\omega/a)t_1}}_{\omega_1} dt_1 = \frac{X(\omega_1)}{|a|} = \frac{1}{|a|} X(\omega/a)$$

Differentiation and Duality Property

$x(t) \Leftrightarrow X(\omega)$. Compute $Y(\omega)$ for $y(t) = \frac{d}{dt} x(t)$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{Synthesis})$$

$$\begin{aligned} \frac{d}{dt} x(t) &= \frac{1}{2\pi} \frac{d}{dt} \left[\int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \right] \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) j\omega e^{j\omega t} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \underbrace{(j\omega X(\omega))}_{Y(\omega)} e^{j\omega t} d\omega \end{aligned}$$

$$y(t) \Leftrightarrow Y(\omega) = j\omega X(\omega)$$

$x(t) \Leftrightarrow X(\omega)$. Then $x(t) \Leftrightarrow 2\pi X(-\omega)$ $\xrightarrow{t \leftrightarrow \omega}$ $\omega \leftrightarrow t$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (\text{Synthesis})$$

$$x(-t_1) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{-j\omega t_1} d\omega \quad (t_1 = -t)$$

$$2\pi x(-t) = \int_{-\infty}^{\infty} X(\omega) e^{-j\omega t} d\omega = \text{FT of } X(\omega).$$

Some more examples: DC Signal, Complex Exponential, Sinusoidal signal

Signal ①:- $x(t) = 1 \quad \forall t$

By using duality Prop

$$\delta(t) \Leftrightarrow 1$$

$$1 \Leftrightarrow 2\pi \underbrace{\delta(-\omega)}_{\text{impulse at '0'}}$$

Signal ②:- $x(t) = e^{-j\omega t}$

[Note: A Complex signal]

$$X(\omega) = e^{-j\omega \tau}$$

Signal ③:- $x(t) = \cos(\omega t)$

$$= \frac{1}{2} (e^{j\omega t} + e^{-j\omega t})$$

$$X(\omega) = \frac{1}{2}$$

$$FT(x_1 + x_2) = FT(x_1) + FT(x_2)$$

To be
discussed
in the
next class

Area under the Curve and Conditions for Convergence

Convergence :- Point wise Convergence :- $\hat{x}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) e^{j\omega t} d\omega$

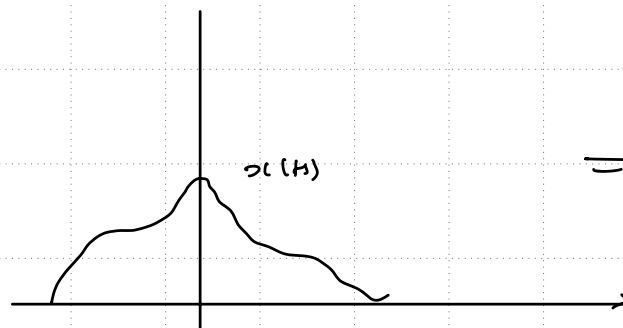
$\hat{x}(t) \approx x(t)$ at all t ←

Mean Square error Sense :- $\hat{x}(t)$ & $x(t)$ have the same prop

$$\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty.$$

Area under the Curve & its relation to $x(0)$:-

①



⇒ area under the signal?

$$\begin{aligned} x(\omega) \Big|_{\omega=0} &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\ &= \text{area under the Curve} \end{aligned}$$